

Scoped Evidence Draft for Exact Dense PVW/MAT-SAB

Anonymous

Abstract

We evaluate an exact dense PVW/MAT-RLWE realization of 2025/686-style sparse amortized bootstrapping [4]. The accumulator uses one shared mask with r independent body lanes, and the primary endpoint is complete bootstrapping time per processed plaintext lane, $T_{\text{bootstrap}}/r$, against r repeated scalar SAB executions under the same backend. On the selected binary rows in Table 1, the current implementation records mean speedups from 1.256900x to 1.357500x, with the weakest reported 95% CI lower bound 1.229193x. The same rows record zero PVW/scalar/pair final-output failures in 20 deterministic seeds per row. The result is deliberately scoped: it is not a non-binary claim, not a compact-selector construction, not a broad novelty claim, and not a theoretical optimality theorem.

1 Research Question and Metric

The research question is whether replacing r independent RLWE/SAB executions with an r -body MAT-RLWE accumulator improves the full bootstrapping cost per handled plaintext lane. The comparison is therefore not a single PVW/MAT run against a single scalar run. The measured speedup is

$$S_r = \frac{T_{\text{scalar}, r \text{ repeated}}}{T_{\text{PVW/MAT}, r \text{ lanes}}},$$

which is equivalent to comparing amortized $T_{\text{bootstrap}}/r$ for the same r -lane workload. All performance rows in this draft use complete-SAB timing and keep the scalar SAB path as the reference implementation.

2 Related Work Boundary

The target baseline is the 2025/686 SAB construction [4]. Incomplete-NTT acceleration is adjacent post-686 transform/backend work and is not folded into the PVW/MAT lane-batching claim [11]. Shared-mask packed-message TFHE creates prior-art risk for broad shared-mask novelty language [1]. PVW packing and packed ciphertext background are treated as building blocks [2], and TFHE/external-product terminology is background rather than a contribution [3]. Amortized, batched, and SIMD bootstrapping axes are established related work [9], [5], [10], [8], [6], and [7].

3 Algorithm Object

The implemented object is exact dense PVW/MAT-SAB. The scalar state has one mask/body pair per independent lane. The PVW/MAT state keeps one shared mask and r body lanes. In this implementation, r denotes independent LUT/SAB lanes, not adjacent accumulator-index packing. The correctness invariant is lane-wise: after each promoted stage and in the final output, PVW/MAT body lane q must match the scalar SAB reference for lane q under the same input, LUT, key material, backend, and parameter row.

The exact dense path preserves the scalar SAB semantics and amortizes shared mask/schedule work, but it also pays dense MAT external-product and resource costs. The current evidence therefore supports a measured complete-SAB throughput claim on selected binary rows only. It does not prove that dense MAT is asymptotically or universally optimal among all possible SAB-compatible MAT-RLWE layouts.

Table 1: Selected binary complete-SAB throughput and side-condition evidence. The speedup is the repeated-scalar total time divided by the PVW/MAT total time for the same r-lane workload, equivalently a comparison of amortized $T_{\text{bootstrap}}/r$.

Param	r	Mean	CI low	CI high	Failures	Key bytes	Keygen	RSS
SET_4_5_2048	4	1.357500	1.344091	1.370909	0/0/0	1.069425	1.338877	1.000880
SET_4_5_2048	2	1.264700	1.255514	1.273886	0/0/0	1.014389	1.262692	0.991062
SET_2_3_4096	2	1.256900	1.229193	1.284607	0/0/0	1.006136	1.102149	0.982190
SET_2_3_4096	4	1.329600	1.317739	1.341461	0/0/0	1.031844	1.357920	0.968173

4 Implementation Scope

The implementation keeps the original scalar SAB path available for A/B testing and routes the optimized path through explicit PVW/MAT entry points. The paper-facing implementation description should name the exact dense PVW/MAT path, active-buffer/copyback reductions where they are part of the measured code path, and the AVX512/spqlios backend used for the selected rows. Backend counter evidence may explain a mechanism, but complete-SAB $T_{\text{bootstrap}}/r$ remains the primary endpoint.

5 Evaluation Summary

Table 1 reports four selected binary rows. Each row uses 10 complete-SAB A/B timing samples and 20 deterministic noise/correctness seeds. The key-size ratio is at most 1.069425 and the keygen-time ratio is at most 1.357920 in this selected matrix, so throughput is reported together with public-key and setup side costs rather than alone.

6 Claim Audit

The allowed claim is narrow: exact dense PVW/MAT-SAB improves complete-SAB $T_{\text{bootstrap}}/r$ over repeated scalar SAB on the selected binary rows under the recorded backend and run protocol. The draft must not claim all-parameter coverage, non-binary coverage, compact selector support, first shared-mask batching, first PVW packing, or theoretical optimality. These blocked statements are tracked in `claim_audit.csv`.

7 Limitations and Next Gates

This draft is an evidence artifact, not a final submission. Remaining gates include venue/template compilation, unresolved bibliography rows, any refreshed native-counter attribution needed for final wording, and separate proofs or experiments for non-binary, compact, or theoretically optimal MAT-RLWE SAB variants.

References

- [1] Loris Bergerat, Charlotte Bonte, Benjamin R. Curtis, Jean-Baptiste Orfila, Pascal Paillier, and Samuel Tap. Sharing the mask: TFHE bootstrapping on packed messages. *IACR Trans. Cryptogr. Hardw. Embed. Syst.*, 2025(4):925–971, 2025.
- [2] Zvika Brakerski, Craig Gentry, and Shai Halevi. Packed ciphertexts in lwe-based homomorphic encryption. In Kaoru Kurosawa and Goichiro Hanaoka, editors, *Public-Key Cryptography - PKC 2013 - 16th International Conference on Practice and Theory in Public-Key Cryptography, Nara, Japan, February 26 - March 1, 2013. Proceedings*, volume 7778 of *Lecture Notes in Computer Science*, pages 1–13. Springer, 2013.
- [3] Ilaria Chillotti, Nicolas Gama, Mariya Georgieva, and Malika Izabachène. TFHE: fast fully homomorphic encryption over the torus. *J. Cryptol.*, 33(1):34–91, 2020.

- [4] Antonio Guimarães and Hilder V. L. Pereira. Fast amortized bootstrapping with small keys and polynomial noise overhead. In Chun-Ying Huang, Jyh-Cheng Chen, Shiuh-Pyng Shieh, David Lie, and Véronique Cortier, editors, *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security, CCS 2025, Taipei, Taiwan, October 13-17, 2025*, pages 2967–2981. ACM, 2025.
- [5] Antonio Guimarães, Hilder V. L. Pereira, and Barry Van Leeuwen. Amortized bootstrapping revisited: Simpler, asymptotically-faster, implemented. In Jian Guo and Ron Steinfeld, editors, *Advances in Cryptology - ASIACRYPT 2023 - 29th International Conference on the Theory and Application of Cryptology and Information Security, Guangzhou, China, December 4-8, 2023, Proceedings, Part VI*, volume 14443 of *Lecture Notes in Computer Science*, pages 3–35. Springer, 2023.
- [6] Feng-Hao Liu and Han Wang. Batch bootstrapping I: - A new framework for SIMD bootstrapping in polynomial modulus. In Carmit Hazay and Martijn Stam, editors, *Advances in Cryptology - EUROCRYPT 2023 - 42nd Annual International Conference on the Theory and Applications of Cryptographic Techniques, Lyon, France, April 23-27, 2023, Proceedings, Part III*, volume 14006 of *Lecture Notes in Computer Science*, pages 321–352. Springer, 2023.
- [7] Feng-Hao Liu and Han Wang. Batch bootstrapping II: - bootstrapping in polynomial modulus only requires $\tilde{o}(1)$ FHE multiplications in amortization. In Carmit Hazay and Martijn Stam, editors, *Advances in Cryptology - EUROCRYPT 2023 - 42nd Annual International Conference on the Theory and Applications of Cryptographic Techniques, Lyon, France, April 23-27, 2023, Proceedings, Part III*, volume 14006 of *Lecture Notes in Computer Science*, pages 353–384. Springer, 2023.
- [8] Zeyu Liu and Yunhao Wang. Amortized functional bootstrapping in less than 7 ms, with $\tilde{o}(1)$ polynomial multiplications. In Jian Guo and Ron Steinfeld, editors, *Advances in Cryptology - ASIACRYPT 2023 - 29th International Conference on the Theory and Application of Cryptology and Information Security, Guangzhou, China, December 4-8, 2023, Proceedings, Part VI*, volume 14443 of *Lecture Notes in Computer Science*, pages 101–132. Springer, 2023.
- [9] Daniele Micciancio and Jessica Sorrell. Ring packing and amortized FHEW bootstrapping. In Ioannis Chatzigiannakis, Christos Kaklamanis, Daniel Marx, and Donald Sannella, editors, *45th International Colloquium on Automata, Languages, and Programming, ICALP 2018, Prague, Czech Republic, July 9-13, 2018*, volume 107 of *LIPICs*, pages 100:1–100:14. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2018.
- [10] Gabrielle De Micheli, Duhyeong Kim, Daniele Micciancio, and Adam Suhl. Faster amortized FHEW bootstrapping using ring automorphisms. In Qiang Tang and Vanessa Teague, editors, *Public-Key Cryptography - PKC 2024 - 27th IACR International Conference on Practice and Theory of Public-Key Cryptography, Sydney, NSW, Australia, April 15-17, 2024, Proceedings, Part IV*, volume 14604 of *Lecture Notes in Computer Science*, pages 322–353. Springer, 2024.
- [11] Thales B. Paiva, Gabrielle De Micheli, Syed Mahbub Hafiz, Marcos A. Simplicio Jr., and Bahattin Yildiz. Faster amortized bootstrapping using the incomplete NTT for free. *IACR Trans. Cryptogr. Hardw. Embed. Syst.*, 2025(4):520–543, 2025.